

review

Def.

$$1) \int k dx = kx + C.$$

P.f.

$$\frac{d}{dx}(kx + C) = k \quad \square$$

$$2) \int x^n dx = \frac{1}{n+1} x^{n+1} + C$$

P.f.

$$\frac{d}{dx} \left(\frac{1}{n+1} x^{n+1} + C \right) = \frac{1}{n+1} \cdot (n+1) x^{(n+1)-1} = x^n \quad \square$$

$$3) \int k f(x) dx = k \int f(x) dx.$$

P.f.

$$\frac{d}{dx} \left(k \int f(x) dx \right) = k \cdot \frac{d}{dx} \left(\int f(x) dx \right) = k \cdot f(x) \quad \square$$

$$4) \int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$$

P.f.

$$\begin{aligned} \frac{d}{dx} \left(\int f(x) dx \pm \int g(x) dx \right) &= \frac{d}{dx} \left(\int f(x) dx \right) \pm \frac{d}{dx} \left(\int g(x) dx \right) \\ &= f(x) \pm g(x) \quad \square \end{aligned}$$